

MACHINE LEARNING

SYLLABUS

NGO QUOC VIET-2025

CONTENTS

- Syllabus
- Course Learning Outcomes
- Programs
- Evaluation structures.
- Books

LEARNING OUTCOMES

O	CLO	
O1	CLO1	Use Python well.
O2	CLO2	Understand, choose, evaluate suitable machine learning algorithms.
	CLO3	Implemen Clustering (s), kNN, ANN, RNN, Naïve Bayes, DT, Boosting.
O3	CLO4	Use the machine learnine packages to solve real problems.
O4	CLO5	Understand machine learning's roles in AI.

MAIN TOPICS

- Introduction
- Data visualization
- Supervised vs unsupervised, parametric vs nonparametric learnings.
- Regression algorithms
- Clustering algorithms (kNN).
- kNN, Decision trees, and boosting algorithms.
- Naïve Bayes algorithm.
- ANN, RNN, and LSTM.
- Support vector machine.

SCORE STRUCTURE

- Midterm and assignments: 50%
- Final projects: 50%. Can be Proposed or Recommendation Projects.

BOOKS AND TOOLS

1. Giuseppe Bonaccorso (2018). Machine Learning Algorithms, *2nd Edition*. Packt.
2. Mettu Sriniva, G. Sucharitha, Anjanna Matta (2021). Machine Learning Algorithms and Applications. Wiley
3. Suhas Pote (2023). Machine Learning in Production. BPB.
4. Christian Kästner (2025). Machine Learning in Production: From Models to Products. MIT Press. https://mlip-cmu.github.io/book/